

Bryan M. Maitland

Curriculum Vitae
July 2026

Rocky Mountain Research Station,
USDA Forest Service, Boise, ID, USA
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Employment

2024–Present **Research Fish Biologist**, Rocky Mountain Research Station, USDA Forest Service
2023–2024 **Research Associate**, Dept. of Civil and Environmental Engineering, University of Wisconsin-Madison
2022–2023 **Postdoctoral Fellow**, Center for Limnology, University of Wisconsin-Madison
2020–2022 **WI Water Resources Postdoctoral Fellow**, Aquatic Sciences Center (UW-Madison) and WDNR

Education

2020 Ph.D. Ecology University of Wyoming
2015 M.Sc. Conservation Biology (Fisheries) University of Alberta
2011 B.S. Conservation Biology *cum laude* SUNY: College of Environmental Science and Forestry

Research Summary

- Professional expertise: Conservation ecology of freshwater fishes using stable isotopes, spatial analysis, and predictive modeling to assess population and food web responses to climate and land use change.
- Since 2014 I have authored 25 papers in peer-reviewed journals (pages 1–2).
- On Google Scholar my h-index is 13 with 708 total citations (as of 16 July 2026).
- I have authored 1 R package as a result of my research with a total of 3139 downloads (page 2).
- I have contributed 47 papers and posters at AFS, SFS, ASLO, AGU, IAGLR, and regional chapter meetings (pages 6–8).

Peer-Reviewed Publications

* indicates shared first authorship. Underlined names indicate undergraduate research mentees.

25. Keppeler, F.W., Giarrizzo, T., Montăna, C.G., Adite, A., Akin, Ş., Angelini, R., **Maitland, B.M.**, Arantes, C.C., Benedito, E., Bokhutlo, T., El-Sabaawi, R., Garcia, A., Gonzalez-Bergonzoni, I., Hoeinghaus, D., Hoffman, J., ... Winemiller, K.O. (2026). What determines trophic niche breadth? A global analysis of freshwater fishes using isospaces. *Reviews in Fish Biology and Fisheries*, 36(13). <https://doi.org/10.1007/s11160-025-10023-2>
24. Sparks*, M.M., **Maitland***, B.M., Felts, E., Swartz, A., & Frater, P. (2026). hatchR: A toolset to predict when fish hatch and emerge. *Fisheries*, 51(2), 49–59. <https://doi.org/10.1093/fshmag/vuaf078>
23. Barrus, N.T., **Maitland, B.M.**, & Rahel, F.J. (2024). Assessing a standardized method to identify optimal baselines for trophic position estimation in stable isotope studies of stream ecosystems. *Hydrobiologia*, 851, 4673–4691. <https://doi.org/10.1007/s10750-024-05618-y>
22. **Maitland, B.M.**, Bootsma, H.A., Bronte, C.R., Bunnell, D.B., Feiner, Z.S., Fenske, K.H., Fetzer, W.W., Foley, C.J., Gerig, B.S., Happel, A., Höök, T.O., Keppeler, F.W., Kornis, M.S., Lepak, R.F., McNaught, A.S., ... Jensen, O.P. (2024). Testing food web theory in a large lake: The role of body size in habitat coupling in Lake Michigan. *Ecology*, 105(10), e4413. <https://doi.org/10.1002/ecy.4413>
21. **Maitland, B.M.**, & Rahel, F.J. (2023). Aquatic food web expansion and trophic redundancy along the Rocky Mountain—Great Plains ecotone. *Ecology*, 104(7), e4103. <https://doi.org/10.1002/ecy.4103>
20. Ward, N.K., Lynch, A.J., Beever, E.A., Booker, J., Bouska, K.L., Embke, H., Houser, J.N., Kocik, J.F., Kocik, J., Lawrence, D.J., Lemon, M.G., Limpinsel, D., Magee, M.R., **Maitland, B.M.**, McKenna, O., ... Wilkening, J.L. (2023). Reimagining large river management using the Resist–Accept–Direct (RAD) framework in the Upper Mississippi River. *Ecological Processes*, 12(1), 48. <https://doi.org/10.1186/s13717-023-00460-x>
19. Feiner, Z.S., Shultz, A.D., Sass, G.G., Trudeau, A., Mitro, M.G., Dassow, C.J., Latzka, A.W., Isermann, D.A., **Maitland, B.M.**, Homola, J.J., Embke, H.S., & Preul, M. (2022). Resist-accept-direct (RAD) considerations for climate change adaptation in fisheries: The Wisconsin experience. *Fisheries Management and Ecology*, 29(4), 346–363. <https://doi.org/10.1111/fme.12549>
18. Kirk, M.A., **Maitland, B.M.**, Hickerson, B.T., Walters, A.W., & Rahel, F.J. (2022). Climatic drivers and ecological impacts of a rapid range expansion by non-native smallmouth bass. *Biological Invasions*, 24, 1311–1326. <https://doi.org/10.1007/s10530-021-02724-z>
17. Lapidés*, D.A., **Maitland***, B.M., Zipper, S.C., Latzka, A.W., Pruitt, A., & Greve, R. (2022). Advancing environmental flows approaches to streamflow depletion management. *Journal of Hydrology*, 607, 127447. <https://doi.org/10.1016/j.jhydrol.2022.127447>
16. **Maitland, B.M.**, & Latzka, A.W. (2022). Shifting climate conditions affect recruitment in Midwestern stream trout, but depend on seasonal and spatial context. *Ecosphere*, 13(12), e4308. <https://doi.org/10.1002/ecs2.4308>

15. Medinski, N.A., **Maitland, B.M.**, Jardine, T.D., Drake, D.A.R., & Poesch, M.S. (2022). A catastrophic coal mine spill in the Athabasca River watershed induces isotopic niche shifts in stream biota including an endangered rainbow trout ecotype. *Canadian Journal of Fisheries and Aquatic Sciences*, 1–14. <https://doi.org/10.1139/cjfas-2021-0112>
14. **Maitland, B.M.**, Martínez del Río, C., & Rahel, F.J. (2021). Effect of temperature on ^{13}C and ^{15}N incorporation rates and discrimination factors in two North American fishes. *Canadian Journal of Fisheries and Aquatic Sciences*, 78(12), 1833–1840. <https://doi.org/10.1139/cjfas-2021-0057>
13. **Maitland, B.M.**, & Rahel, F.J. (2021). Nonlethal fin sampling of North American freshwater fishes for food web studies using stable isotopes. *North American Journal of Fisheries Management*, 41(2), 410–420. <https://doi.org/10.1002/nafm.10539>
12. Voter, C.B., Guerrero-Bolaño, F.J., Latzka, A.W., **Maitland, B.M.**, & Hauxwell, J.A. (2021). Adaptable university-agency early-career fellowship program creates a win-win-win for Wisconsin's waters. *Journal of Contemporary Water Research and Education*, 174, 140–155.
11. Kirk, M.A., **Maitland, B.M.**, & Rahel, F.J. (2020). Spatial scale, reservoirs and nonnative species influence the homogenization and differentiation of Great Plains—Rocky Mountain fish faunas. *Hydrobiologia*, 847, 1–15. <https://doi.org/10.1007/s10750-019-04129-5>
10. Alston, J.M., **Maitland, B.M.**, Brito, B.T., Esmaeili, S., Ford, A.T., Hays, B., Jesmer, B.R., Molina, F.J., & Goheen, J.R. (2019). Reciprocity in restoration ecology: When might large carnivore reintroduction restore ecosystems? *Biological Conservation*, 234, 82–89. <https://doi.org/10.1016/j.biocon.2019.03.021>
9. Hickerson, B.T., **Maitland, B.M.**, & Walters, A.W. (2019). Effects of multiple nonnative fish on an imperiled cyprinid, Hornyhead Chub *Nocomis biguttatus*. *Transactions of the American Fisheries Society*, 148(6), 1132–1145. <https://doi.org/10.1002/tafs.10203>
8. **Maitland, B.M.**, O'Malley, B.P., & Stewart, D.J. (2019). Subsurface water piping transports plankton and prevents meromixis in a deep volcanic crater lake (Dominica, West Indies). *Hydrobiologia*, 839(1), 119–130. <https://doi.org/10.1007/s10750-019-04000-7>
7. Walker, R.H., **Maitland, B.M.**, LaSharr, T.N., Rosing, M.N., & Ben-David, M. (2018). Fate of juvenile salmonids stranded in off-channel pools: Implications for nutrient transfers. *Aquatic Sciences*, 80(1), 10. <https://doi.org/10.1007/s00027-017-0562-z>
6. Craig, L.S., Olden, J.D., Arthington, A.H., Entekhabi, S., Hawkins, C.P., Kelly, J.J., Kennedy, T.A., **Maitland, B.M.**, Rosi, E.J., Roy, A.H., Strayer, D.L., Tank, J.L., West, A.O., & Wooten, M.S. (2017). Meeting the challenge of interacting threats in freshwater ecosystems: A call to scientists and managers. *Elementa: Science of the Anthropocene*, 5(72), 1–15. <https://doi.org/10.1525/elementa.256>
5. Pandit, S.N., **Maitland, B.M.**, Pandit, L.K., Poesch, M.S., & Enders, E.C. (2017). Climate change risks, extinction debt, and conservation implications for a threatened freshwater fish: Carmine shiner (*Notropis percobromus*). *Science of the Total Environment*, 598, 1–11. <https://doi.org/10.1016/j.scitotenv.2017.03.228>
4. **Maitland, B.M.**, Poesch, M., & Anderson, A.E. (2016a). Prioritising culvert removals to restore habitat for at-risk salmonids in the boreal forest. *Fisheries Management and Ecology*, 23(6), 489–502. <https://doi.org/10.1111/fme.12188>
3. **Maitland, B.M.**, Poesch, M., Anderson, A.E., & Pandit, S.N. (2016b). Industrial road crossings drive changes in community structure and instream habitat for freshwater fishes in the boreal forest. *Freshwater Biology*, 61, 1–18.
2. **Maitland, B.M.**, Cooke, S.J., & Poesch, M. (2015). Finding the path to a successful graduate and research career: Advice for early career researchers. *Fisheries*, 40(8), 399–403. <https://doi.org/10.1080/03632415.2015.1065253>
1. O'Malley, B.P., & **Maitland, B.M.**, (2014). Observations on the freshwater zooplankton of Dominica, West Indies, including records from the former Matthieu landslide-dam lake. *Pan-American Journal of Aquatic Sciences*, 9(3), 228–333.

R-packages

1. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P.N. (2025). *hatchR: Predict Fish Hatch and Emergence Timing*. <https://github.com/bmait101/hatchR>

Technical Reports

3. **Maitland, B.M.**, & Loheide, S.P. (2025). *Pumped storage hydropower in California: Environmental effects and opportunities for co-benefits*. University of Wisconsin-Madison.
2. Mitro, M., Feiner, Z.S., Latzka, A.W., Embke, H.S., Homola, J.J., Isermann, D.A., **Maitland, B.M.**, Sass, G.G., Shaw, S., Shultz, A.D., Tsyehaye, I., & Vander Zanden, J. (2021). *Wisconsin Initiative on Climate Change Impacts: Fisheries Working Group Report*. Nelson Institute for Environmental Studies, University of Wisconsin-Madison.
1. Roseman, E.F., Riley, S.C., Farha, S.A., **Maitland, B.M.**, Tucker, T.R., Provo, S.A., & McLean, M.W. (2013). *Status and Trends of the Lake Huron Offshore Demersal Fish Community, 1976–2012*. USGS Great Lakes Science Center.

Theses

2. **Maitland, B.M.**, (2020). *Isotopic Ecology of Aquatic Communities Along the Rocky Mountains—Great Plains Ecotone* [PhD thesis]. University of Wyoming.

1. **Maitland, B.M.**, (2015). *Stream Crossings in the Western Boreal Forest: Assessing Impacts and Prioritizing Restoration for Native Freshwater Fishes* [Master's thesis]. University of Alberta.

Manuscripts Under Consideration

2. Sparks, M.M., Leavell, B.C., & **Maitland, B.M.**, (under review). A generalizable tool for predicting developmental phenology for wild poikilotherms. *Ecology and Evolution*.
1. Vagnon, C., Jardine, T.D., Hayden, B., Tunney, T.D., Sanchez-Hernandez, J., McIntosh, A.A., Alves, C.B., Arostegui, M.C., Baudry, T., Beesley, L., Belliard, J., Bellouard, N., Boulêtreau, S., Britton, R., **Maitland, B.M.**, ... Cucherousset, J. (under review). IsoFreshID: A global database of individual-level $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ stable isotopes of freshwater food webs. *Scientific Data*.

Unpublished Working Papers

2. Hargrove, J., **Maitland, B.M.**, Sparks, M.M., & Isaak, D.J. (2026). *From thermal barriers to connectivity corridors: Shifting drivers of Rainbow Trout introgression into Westslope Cutthroat Trout over two decades*.
1. **Maitland, B.M.**, Thurow, R.F., Sparks, M.M., Isaak, D.J., & Buffington, J.M. (2026). *Habitat heterogeneity and phenotypic diversity: The influence of stream attributes on timing of Chinook Salmon spawning*. https://github.com/morgan-sparks/mfsr_phenology

Grants

Since 2015, I have led, co-led, or partnered on \$0.3 million in funded projects and contributed to multiple invited proposals exceeding \$2 million in scope. Funded and pending grants > \$10,000 are given below:

Pending	Lead-PI. "Idaho Bull Trout food web ecology and hybrid interactions". <i>Funding from U.S. Fish and Wildlife Service</i> . Collaborators: M. Sparks (Co-PI).	\$20,000
Pending	Co-PI. "Idaho Bull Trout migration timing and habitat use". <i>Funding from U.S. Fish and Wildlife Service</i> . Collaborators: M. Sparks (Lead-PI), B. Bowersox.	\$159,900
2026–2028	Co-PI. "From flames to fish: Co-management for wildfire and aquatic species in Idaho". <i>Funding from Joint Fire Science Program</i> . Collaborators: J. Peirce (Lead-PI), B. Sadegh, A. Bergstrom, C. Wasler, M. Sparks, D. Isaak.	\$150,284
2025–2027	Lead-PI. "Monitoring Chinook Salmon spawning from space". <i>Funding from Trout Unlimited</i> . Collaborators: D. Daulwater, M. Sparks.	\$38,122
2024–2027	Lead-PI. "Chinook Salmon life history and otolith microchemistry". <i>Funding from UC-Davis and USDA Forest Service</i> . Collaborators: C. Jeffries.	\$50,184
2020–2021	Lead-PI. "Accessible underwater landscapes for non-STEM audiences". <i>Funding from UWyo Biodiversity Institute</i> . Collaborators: F. Rahel.	\$19,866
2019–2020	Lead-PI. "Diet specialization along stream gradients". <i>Funding from UWyo Zoology</i> . Collaborators: F. Rahel.	\$22,398
2015–2018	Lead-PI. "Factors that influence the diversity of fish assemblages in Wyoming". <i>Funding from UWyo Biodiversity Institute</i> . Collaborators: F. Rahel.	\$25,207

Invited full proposals that were withdrawn due to federal funding issues in the U.S.:

2025	Lead-PI. "A palaeoecological toolset to address shifting baseline syndrome". <i>Funding from DoD SERDP</i> . Collaborators: M. Sparks, B. Finney, K. Carim, A. Fraik, R. Thurow, C. Shobe, D. White.	\$1,500,000
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Invited full proposals that were not awarded:

2023	Lead-PI. "Cross-lake comparison of habitat coupling in Lakes Michigan and Superior". <i>Funding from WI Sea Grant</i> . Collaborators: O. Jensen.	\$258,571
2021	Lead-PI. "Climate change and groundwater pumping impacts on Wisconsin trout fisheries". <i>Funding from USGS MW CASC</i> . Collaborators: A. Latzka, O. Jensen, S. Loheide.	\$399,853

Honors and Awards

Professional Awards

- 2026 Outstanding Service Award, U.S. Forest Service, Rocky Mountain Research Station
- 2024 Outstanding Service Award, U.S. Forest Service, Rocky Mountain Research Station
- 2021 Best Professional Presentation Award, Wisconsin Chapter American Fisheries Society

Fellowships

- 2020 Wisconsin Water Resources Postdoctoral Fellow, University of Wisconsin-Madison
- 2019 Dean's Graduate Fellowship, College of Arts and Sciences, University of Wyoming

Student Awards

- 2018 Eugene Maughan Graduate Award, Western Division American Fisheries Society
- 2018 Hank Gardner Graduate Award, Department of Zoology and Physiology University of Wyoming
- 2017 Clemens-Rigler Award, Canadian Aquatic Resources Section American Fisheries Society
- 2017 Board of Visitors Award, College of Arts and Sciences University of Wyoming
- 2017 Dennis Anderson Fisheries Award, Department Zoology and Physiology University of Wyoming
- 2017 William Trachtenberg Scholarship, Sustainable Fisheries Foundation
- 2017 Vern Bressler Fisheries Award, Department of Zoology and Physiology University of Wyoming
- 2015 Excellence in Fisheries Award, Canadian Aquatic Resources Section American Fisheries Society
- 2015 Clemens-Rigler Award, Canadian Aquatic Resources Section American Fisheries Society

Professional Service and Activities

Service, Teams, and Committees

- 2025–2029 Mentor of Practice, NSF IRES: Climate Change Resilience in an Aquatic Bright Spot, Jensen (PI), UW-Madison
- 2021–2022 Member, Fisheries Working Group, Wisconsin Initiative on Climate Change Impacts
- 2020–2022 Member, Trout Species Team, Wisconsin Department of Natural Resources
- 2016–2020 Social Media Coordinator, Program in Ecology, University of Wyoming
- 2016–2017 President, Program in Ecology, University of Wyoming
- 2015–Present Member, Freshwater Working Group, Society for Conservation Biology

Membership in Professional Societies

- 2019–2023 Galápagos Conservancy
- 2016–Present Society for Freshwater Science
- 2016–2019 Association for the Science of Limnology and Oceanography
- 2015–Present Ecological Society of America
- 2010–Present American Fisheries Society
- 2010–Present Society for Conservation Biology
- 2010–2016 International Association of Great Lakes Research

Service in Professional Societies

- 2026 Program co-chair, Fish and Fire, American Fisheries Society Annual Meeting
- 2024 Program co-chair, Science to Restore and Manage Wildland Aquatic Habitats and Watersheds, AGU Fall Meeting
- 2018 Excom Member, CO-WY State Chapter, American Fisheries Society
- 2016 President, UWyo Student Chapter, American Fisheries Society
- 2016 Excom Member, CO-WY State Chapter, American Fisheries Society
- 2016 Program co-chair, Western Division American Fisheries Society Student Colloquium

Demonstrations, Short Courses, and Training Sessions

- 2023–Feb “Programming with R”. Wisconsin Chapter AFS Annual Meeting, Stevens Point, WI.
- 2022–Feb “Data visualization in R”. Wisconsin Chapter AFS Annual Meeting, Virtual.

Leadership development

- 2021 Equity and Inclusion Workshop EQT By Design via WI Sea Grant
- 2008 Leave No Trace Master Educator Center for Outdoor Ethics, NOLS

Peer-Reviewing Activities

- Manuscript Review
 - Since 2016, I have reviewed 58 manuscripts from 18 journals (~ 5.8 per year).
 - Journals include: *Aquatic Invasions* (2), *Canadian Journal of Fisheries and Aquatic Sciences* (2), *Diversity and Distributions*, *Transportation Research: Part D* (1), *Ecological Applications* (4), *Ecosphere* (5), *Ecology* (3), *Environmental Biology of Fishes* (2), *Fisheries* (5), *Fisheries Research* (2), *Freshwater Biology* (3), *Freshwater Science*

(4), *Frontiers in Ecology and Evolution* (2), *Global Change Biology* (2), *Hydrobiologia* (8), *Journal of Ecology* (1), *Journal of Great Lakes Research* (2), *North American Journal of Fisheries Management* (5), *Oecologia* (1), *PeerJ* (1), *River Research and Applications* (1), *Restoration Ecology* (2).

➤ External Grants

- Lake Champlain Fisheries Research Program.
- NSF, Division of Environmental Biology (DEB).

➤ Internal Review

- I provide 5-10 internal reviews of manuscripts for USFS and USGS colleagues annually.

University Involvement

Adjunct and affiliate appointments

- 2026-present: Graduate Faculty, Department of Biology, Geology, and Environmental Science, University of Tennessee at Chattanooga, Chattanooga, TN.

Doctoral and master student committees

- 2026-present: Will M, University of Tennessee at Chattanooga (MS committee member)

Invited Academic Talks

- 2020-May "Exploring isotopic niches in Rocky Mountain Great Plains stream fish". Department of Zoology and Physiology, University of Wyoming.
- 2022-Mar "Coldwater fish in a changing world: Actionable ecology for management and policy". Department of Ecology, Montana State University.
- 2023-Dec "Freshwater science in a changing world: fisheries research for conservation and management". School of Freshwater Science, UW-Milwaukee.
- 2025-Feb "Freshwater science for tomorrow: advancing fish ecology, conservation, and management". Wildlife Biology Program, University of Montana.
- 2026-Feb "Stream fish ecology and conservation under environmental change". Department of Fish, Wildlife, and Conservation Biology, Colorado State University.
- 2026-Mar "From co-production to decision support: Using ecological science to inform fisheries management". Cornell Biological Field Station, Natural Resources and the Environment, Cornell University.

Mentoring

Graduate Students

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|--------------|-----------------|--------------------------------------|
| 2025-Present | Garrison Ferone | University of Idaho |
| 2025-Present | Quinn Smith | University of Wisconsin-Madison |
| 2023-2025 | Dylan Keisser | University of Tennessee, Chattanooga |

Undergraduate Students

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|-----------|------------------|--------------------------------------|
| 2024-2025 | Trevor Davis | University of Tennessee, Chattanooga |
| 2023-2024 | Jonathan Martin | University of Wisconsin-Madison |
| 2018-2020 | Nathan Barrus | University of Wyoming |
| 2019-2020 | Russell Brubaker | University of Wyoming |
| 2018-2019 | Reilly Davis | University of Wyoming |
| 2017 | Brynn Hirschman | University of Wyoming |
| 2017 | Austin Jacobson | University of Wyoming |
| 2016 | Shaun Helsper | University of Wyoming |
| 2016-2017 | Travis Allison | University of Wyoming |
| 2016 | Andrew Annear | University of Wyoming |
| 2015 | Jake Werner | University of Wyoming |

Teaching Experience

Instructor of Record

ZOO 4330 Ichthyology University of Wyoming Spring 2020 3 credits 22 students

Teaching Assistant and Lab Instructor

ZOO 4330	Ichthyology	University of Wyoming	Spring 2020	3 credits	22 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2019	3 credits	20 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2018	3 credits	25 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2017	3 credits	19 students
ZOO 4330	Ichthyology	University of Wyoming	Spring 2016	3 credits	18 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2019	3 credits	21 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2018	3 credits	26 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2017	3 credits	22 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2017	3 credits	19 students
ZOO 4310	Fisheries Management	University of Wyoming	Fall 2017	3 credits	23 students
ZOO 4310	Fisheries and Wildlife Management	University of Alberta	Spring 2015	4 credits	27 students
ZOO 4310	Fisheries and Wildlife Management	University of Alberta	Spring 2014	4 credits	29 students

Guest Lectures

2023-Feb	"Stable isotope mixing models in R". RENR 5500, University of Wyoming.
2020-Dec	"Aquatic ecology in Western ecosystems". LIFE 3400, University of Wyoming.
2019-Apr	"Applications of stable isotopes in fisheries management". ZOO 4310, University of Wyoming.

Outreach and Broader Impacts

Press Coverage and Media

2023-Feb	Milwaukee Journal Sentinel	Topic: Climate change and stream trout
2023-Jan	Live Interview: WPR Morning Show	Topic: Climate change and stream trout
2022-Dec	Water Blogged	Topic: Climate change and stream trout
2022-Mar	Trout Unlimited, Madison Chapter	Topic: Climate change and stream trout
2017-Nov	Magazine Article: The Tributary	Topic: Cutthroat trout
2017-Aug	U Wyoming Science Cafe	Topic: Aquatic connectivity
2016-Jun	U Wyoming Science Cafe	Topic: Aquatic connectivity

Community Service and Youth Engagement

2022-2023	Food service volunteer	Ronald McDonald House, Madison, WI
2016-2020	Science outreach and stream ecology field trips	Laramie High School, Laramie, WY
2016-2019	Group leader, Wyoming Bio-Blitz	Biodiversity Institute, University of Wyoming
2016-2018	Big Brother	Big Brothers Big Sisters of Wyoming
2017-2018	Habitat restoration projects	Girl Scouts of America
2016-2017	Science judge	Wyoming State Science Fair

Conference Presentations

Underlined names indicate undergraduate research mentees.

47. **Maitland, B.M.**, Thurow, R.F., Sparks, M.M., Isaak, D.J., & Buffington, J.M. (2026, May). *Network-scale thermal heterogeneity structures a diverse Chinook Salmon spawning portfolio* [Paper]. WD AFS annual meeting, Portland, OR.
46. Valentine, G.P., Shobe, C.M., Alfred, V.S., Lininger, K.B., Meyer, K., **Maitland, B.M.**, & Fairchild, M.P. (2026, May). *Two years of benthic macroinvertebrate community recovery following low-head dam removal in a semi-arid Rocky Mountain stream* [Paper]. SFS annual meeting, Spokane, WA.
45. Walker, R.H., Daugherty, O., Davis, T., Wildman, D., Wright, L., Hunt, D., & **Maitland, B.M.**, (2026a, May). *That's a mouthful: Availability, feeding ecology, and life history variation of stream fishes during periodical cicada emergences* [Paper]. SFS annual meeting, Spokane, WA.
44. Wildman, D., Daugherty, O., Davis, T., Wright, L., Hunt, D., **Maitland, B.M.**, & Walker, R.H. (2026, May). *Cicadas in the stream: Quantifying periodical cicada drift densities in streams* [Poster]. SFS annual meeting, Spokane, WA.
43. Walker, R.H., Daugherty, O., Davis, T., Wildman, D., Wright, L., Hunt, D., & **Maitland, B.M.**, (2026b, March). *Cicadas in the stream: Evaluating the foraging ecology and life history variation of fishes during a rare but massive resource pulse to stream ecosystems* [Paper]. TN chapter AFS annual meeting, Cumberland Mountain State Park, TN.
42. Ferone, G.S., **Maitland, B.M.**, Isaak, D.J., & Caudill, C.C. (2025, November). *Applying spatial stream networks and species distribution models to two decades of Idaho DEQ BURP stream benthic macroinvertebrate samples* [Paper]. SFS pacific northwest chapter annual meeting, Boise, ID.
41. Sparks, M.M., Leavell, B.C., & **Maitland, B.M.**, (2025, November). *A generalizable tool for predicting developmental phenology for freshwater poikilotherms* [Paper]. SFS pacific northwest chapter annual meeting, Boise, ID.
40. Kirk, M.A., & **Maitland, B.M.**, (2025, August). *Trophic overlap between invasive round gobies and native benthic darters in a stream biodiversity hotspot* [Paper]. AFS annual meeting, San Antonio, TX.

39. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P. (2025a, August). *hatchR: A toolset for predicting when fish hatch and emerge* [Poster]. AFS annual meeting, San Antonio, TX.
38. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P. (2025b, May). *hatchR: A toolset for predicting when fish hatch and emerge* [Poster]. Western division AFS annual meeting, Westminster, CO.
37. **Maitland, B.M.**, Sparks, M.M., Felts, E., Swartz, A., & Frater, P. (2025c, April). *hatchR: A toolset for predicting when fish hatch and emerge* [Paper]. Idaho chapter AFS annual meeting, Boise, ID.
36. Davis, T., Wright, L., Keisser, D., **Maitland, B.M.**, & Walker, R.H. (2024, November). *That's a mouthful: Periodical cicadas, fish body size, and feeding ecology* [Poster]. Southeastern fisheries council meeting, Little Rock, AR.
35. Keisser, D., **Maitland, B.M.**, & Walker, R.H. (2024, November). *Quantifying rare allochthonous insect pulses: How many periodical cicadas enter streams?* [Paper]. Southeastern fisheries council meeting, Little Rock, AR.
34. Walker, R.H., Keisser, D., **Maitland, B.M.**, & Hunt, D. (2024, November). *Quantifying the availability, use, and consequences of periodical cicadas in stream food webs* [Paper]. Southeastern fisheries council meeting, Little Rock, AR.
33. Hauxwell, J.A., Voter, C.B., Guerrero-Bolaño, F.J., Latzka, A.W., & **Maitland, B.M.**, (2024, June). *Adaptable university-agency early-career fellowship program creates win-win-win for Wisconsin's waters* [Paper]. IAGLR annual conference, Milwaukee, WI.
32. **Maitland, B.M.**, (2024, June). *Testing food web theory in a large lake: The role of body size in habitat coupling in Lake Michigan* [Paper]. ASLO annual meeting, Madison, WI.
31. **Maitland, B.M.**, & Latzka, A.W. (2024, May). *Shifting climate conditions affect recruitment in Midwestern trout, but depend on seasonal and spatial context* [Paper]. Washington, british columbia, and idaho chapters AFS joint meeting, Spokane, WA.
30. Walker, R.H., & **Maitland, B.M.**, (2024, May). *The cicadian rhythm: Preparation for evaluating the availability, use, and consequences of periodical cicadas to stream fishes* [Paper]. Iowa chapter AFS annual meeting, Iowa City, IA.
29. **Maitland, B.M.**, Jensen, O.P., Hoffman, J.C., & Keppeler, F.W. (2023a, August). *Testing food web theory in large lakes: The role of body size in habitat coupling in Lake Michigan* [Paper]. AFS annual meeting, Grand Rapids, MI.
28. Clancy, N., Isaak, D.J., **Maitland, B.M.**, Budy, P., & Walters, A. (2023, March). *Thermal refuges for climate vulnerable fishes in Wyoming* [Paper]. Colorado/wyoming chapter AFS annual meeting, Laramie, WY.
27. **Maitland, B.M.**, Jensen, O.P., Hoffman, J.C., Keppeler, F.W., Bunnell, D.B., McNaught, A.S., & Gerig, B.S. (2023b, February). *Body size, trophic position, and energy coupling in Lake Michigan* [Paper]. Wisconsin chapter AFS annual meeting, Stevens Point, WI.
26. Sass, G.G., Beard, D., Broda, S., Carpenter, S.R., Chipps, S.R., Cichosz, T.A., Dassow, C.J., Elwer, B., Embke, H.S., Feiner, Z.S., Gaeta, J.W., Hansen, G.J.A., Hansen, J., Hennessy, J., Homola, J.J., ... Zebro, L. (2023, January). *Depensatory recruitment in Wisconsin walleye: Mechanisms and management* [Paper]. Wisconsin DNR office of applied science webinar series, Virtual.
25. Ward, N.K., Lynch, A.J., Beever, E.A., Bouska, K.L., Embke, H.S., Kocik, J.F., Limpinsel, D., Magee, M.R., **Maitland, B.M.**, Morton, J.M., Mueller, J.M., Oliver, D.C., Rantala, H.M., Sass, G.G., Schultz, A., ... Wilkening, J.L. (2022, August). *Using the Resist-Accept-Direct (RAD) framework to reimagine large river management* [Paper]. AFS annual meeting, Spokane, WA.
24. **Maitland, B.M.**, & Latzka, A.W. (2022, May). *Shifting climate conditions affect recruitment in Midwestern stream trout but depend on seasonal and spatial context* [Paper]. Joint aquatic sciences meeting, Grand Rapids, MI.
23. **Maitland, B.M.**, & Latzka, A.W. (2021, November). *Long-term trends of Midwestern stream trout populations in a changing climate, and the effects of seasonal weather* [Paper]. AFS annual meeting, Baltimore, MD.
22. **Maitland, B.M.**, Latzka, A.W., Frater, P., & Mitro, M.G. (2021a, May). *Untangling the effects of hydrologic change on trout in Wisconsin streams* [Paper]. Wisconsin water week, Virtual.
21. Lapides, D.A., **Maitland, B.M.**, & Pruitt, A.H. (2021, March). *Assessing approaches to quantify hydrological alteration on Wisconsin's streams* [Paper]. Wisconsin section AWRA annual meeting, Virtual.
20. **Maitland, B.M.**, Voter, C.B., Guerrero-Bolaño, F.J., Latzka, A.W., & Hauxwell, J.A. (2021b, March). *Tackling Wisconsin's water challenges through UW water science-policy fellowships and agency partnerships* [Paper]. Wisconsin water week, Virtual.
19. **Maitland, B.M.**, Latzka, A.W., Frater, P., & Mitro, M.G. (2021c, February). *Long-term trout trends in Wisconsin and the role of hydrologic change* [Paper]. Wisconsin chapter AFS annual meeting, Virtual.
18. **Maitland, B.M.**, Fetzer, W.W., Collins, S.M., & Rahel, F.J. (2020, March). *Non-lethal approaches for obtaining stable isotope data for diet studies of North American fishes* [Paper]. CO/WY chapter AFS annual meeting, Laramie, WY.
17. Barrus, N.T., **Maitland, B.M.**, & Rahel, F.J. (2019, October). *Identification of an ideal baseline organism for stream food web studies* [Paper]. Joint meeting of the wildlife and american fisheries society, Reno, NV.
16. **Maitland, B.M.**, & Rahel, F.J. (2019, October). *Stream food web expansion in response to changing environmental conditions* [Paper]. Joint meeting of the wildlife and american fisheries society, Reno, NV.
15. **Maitland, B.M.**, & Rahel, F.J. (2018, January). *Niche partitioning in fish assemblages along longitudinal stream gradient* [Poster]. Canadian conference for freshwater fisheries research, Edmonton, AB.
14. **Maitland, B.M.**, & Rahel, F.J. (2017, June). *Isotopic niches and the longitudinal gradient in Rocky Mountain-Great Plains streams* [Paper]. SFS annual meeting, Raleigh, NC.

13. **Maitland, B.M.**, Walker, R.H., LaSharr, T.N., Rosin, M., & Ben-David, M. (2017a, April). *Juvenile salmonids stranded in flood ponds of Alaska* [Paper]. Program in ecology symposium, Laramie, WY.
12. **Maitland, B.M.**, Walker, R.H., LaSharr, T.N., Rosin, M., & Ben-David, M. (2017b, February). *The fate of juvenile salmonids stranded in flood ponds: Implications for nutrient transfers* [Paper]. Utah and Colorado/Wyoming chapter AFS joint meeting, Grand Junction, CO.
11. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2016a, May). *Culverts in forested watersheds: Impacts and restoration methods for native freshwater fishes* [Paper]. SFS annual meeting, Sacramento, CA.
10. Annear, A.A., **Maitland, B.M.**, & Rahel, F.J. (2016, March). *Stable isotope quantification of resource use by crayfish in the Laramie River to inform lotic food web analyses* [Poster]. Colorado/Wyoming chapter AFS annual meeting, Laramie, WY.
9. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2016b, March). *Culverts in forested watersheds: Impacts and restoration methods for native freshwater fishes* [Paper]. Colorado/Wyoming chapter AFS annual meeting, Laramie, WY.
8. Daniels, S.E., Fitzpatrick, T.A., & **Maitland, B.M.**, (2015, January). *Idiosyncratic resource use by Laramie raccoons: Are trash pandas, trash pandas?* [Poster]. Stable isotope ecology symposium, Laramie, WY.
7. Hirschfield, F., Anderson, A.E., **Maitland, B.M.**, Wagner, M., Zaichkowsky, M., & Mottishaw, C. (2014, October). *Integrating science and management to improve sustainability of forested watersheds in Northwestern Canada* [Poster]. SAF national convention, Salt Lake City, UT.
6. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2014a, August). *Stream crossing impacts on freshwater fishes: Implications for management* [Poster]. Forest industry lecture series, Edmonton, AB.
5. **Maitland, B.M.**, O'Malley, B.P., O'Brien, T.P., Armenio, P.M., Watson, N.M., & Roseman, E.F. (2014b, May). *Zooplankton community dynamics in a northern Lake Huron embayment: The influence of water temperature and larval fish abundance* [Paper]. IAGLR annual meeting, Hamilton, ON.
4. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2014c, May). *Stream crossing assessment procedures as a tool for mitigating impacts on freshwater fish* [Poster]. Columbia mountain institute of applied ecology resource roads conference, Nelson, BC.
3. **Maitland, B.M.**, Poesch, M.S., & Anderson, A.E. (2014d, January). *Stream crossing impacts on freshwater fish and in-stream habitat in Alberta* [Paper]. Canadian conference for freshwater fisheries research, Yellowknife, NWT.
2. **Maitland, B.M.**, Farha, S.A., Darnton, R.W., Smith, K.R., & Riley, S.C. (2012, September). *2012 avian botulism environmental sampling in Sleeping Bear Dunes National Lakeshore* [Paper]. Workshop on GLRI project no. 73 and 91 USGS/NPS meeting, Empire, MI.
1. Discenza, J.D., O'Malley, B.P., **Maitland, B.M.**, & Fernando, D. (2011, April). *Analyzing the genetic variation in American Hart's tongue fern* [Poster]. SUNY: ESF spotlight on student's research, Syracuse, NY.